

COMPUTER VISION PROJECT PROPOSAL

ENHANCING MEDULLOBLASTOMA IDENTIFICATION IN MRI IMAGES THROUGH FEATURE PYRAMID NETWORK INTEGRATION WITH INCEPTION AND SELF ATTENTION MECHANISM

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Introduction

Medulloblastoma (MB) is the most common malignant brain tumor in children, constituting nearly 20 percent of all pediatric brain tumors [1]. It represents the most common intracranial primary malignant tumor in children, with consequences like nausea, vomiting, sensory and motor dysfunction, brain herniation, and ultimately death [2]. The gold standard of diagnosing MB is histopathological examination through biopsy or surgical resection, and these two methods are risky procedures and not always feasible to all patients [3], while there is the Magnetic Resonance Imaging (MRI), a non-invasive method for brain tumor evaluation but it comes with an extreme difficulty in interpretation for experienced neuroradiologists [4].

The advancement in deep learning approaches especially Convolutional Neural Networks (CNNs) has shown great prospects for Artificial intelligence augmented medical imaging analysis, with different literature showing the application of CNN-based models as shown in the recent work section. More recently Fang et al [5] proposed InceptionNet architecture which is a hybrid architecture which combines Inception-based multiscale feature extraction with a multi-head self-attention mechanism for binary classification of MB in MRI images, the model achieved a 90.94% accuracy with 92.79% AUC on the test set thereby performing better than several CNN baselines.

Despite the progress achieved by InceptionNet, the architecture applies self-attention on a single, down sampled feature map. As in typical bottom-up CNNs, this makes the deepest features semantically strong but spatially coarse, which can lead to loss of information about very small lesions, weakly contrasted tumors, or subtle boundaries between medulloblastoma and other posterior fossa tumors [5], [6]. Feature Pyramid Networks (FPN) were specifically proposed to address this limitation by building a top-down feature hierarchy with lateral connections, so that all pyramid levels are both semantically rich and spatially detailed [7]. Integrating FPN into the InceptionNet pipeline is therefore a principled way to enhance the model's ability to represent medulloblastomas across a wide range of sizes and appearances.

Related Works

Recent works that have combined convolutional neural networks and attention mechanisms to improve medical imaging classification especially for brain MRI. Some of them are examined below.

T.Sathwik in [8] focuses on using an ensemble CNN model for the detection of medulloblastoma. The authors leverage pre-trained models through transfer learning and fine-tune the networks on their dataset. Instead of relying on a single model, they combine multiple models to improve performance. Ensemble 1 consists of ResNet50, MobileNetV2, and Xception, achieving an accuracy of 93.33%. Ensemble 2 comprises Xception, MobileNetV2, and InceptionV3, and it also achieves an accuracy of 93.33%. Although both ensembles obtained similar accuracy, Ensemble 1 demonstrated better precision.

Building on CNN-based frameworks, Rasheed et al. [9] proposed a CNN-based framework which is augmented with hybrid attention mechanisms for multi-class brain tumor classification on MRI, the approach distinguishes glioma, meningioma, pituitary tumors and normal brain tissues, their architecture uses the convolutional feature extractors using both custom and pretrained architectures like the Xception and ResNet variants with attention modules to enhance the discriminative regions within the feature maps, the attention is applied at high level feature stages to reweight spatial locations and channels so as to steer the network towards the tumor bearing regions of the MRI and away from the background. Their method show good generalization capability and attained an accuracy of 98.33%, some of the limitations recorded are that the model was not tested on a wide range of dataset, and therefore the generalizability of the model remains uncertain, also, even though attention improves the focus on the relevant regions the model is still constrained by the underlying CNN feature hierarchy, which may reduce its sensitivity to very small or low-contrast lesions especially when heavy down-sampling is done.

Another study by Apostolopoulos et al. [10], also introduced an attention based deep convolutional neural network for brain MRI classification and grading, they termed the architecture AFF-VGG19, their approach extends the VGG 19 architecture with attention feature fusion, which combine information from multiple intermediate layers, their main aim was to fuse low level and high level features using attention mechanisms so that both the local and global features of the MRI can contribute to the final prediction. They implemented this architecture by aggregating the feature maps from different depths of the VGG19 backbone and passing through the attention fusion modules where the more informative channels and spatial regions in the images are given more weight, they recorded better performance when compared to the vanilla VGG19 and other CNN models. While the feature fusion across layers is implemented, the architecture does not explicitly analyze or validate how well the model handles lesions of very different sizes or subtle signal changes at the lesion margins. Moreover, the work does not focus on posterior fossa tumors or medulloblastoma specifically, and thus, does not directly address the particular diagnostic difficulties associated with those entities.

Feature Pyramid Network(FPN) was the focus in study [11] where L.Zhu et. al. worked on modifying the standard architecture. In the standard FPN, features from different levels are summed equally, assuming that all layers are equally important. However, low-level features (detailed edges and textures) and high-level features (semantic information) are fundamentally different. Because there is no adaptive weighting mechanism, this simple summation can lead to poor feature fusion. To address this issue, the paper proposed an Improved Feature Pyramid Network (ImFPN), which fuses multi-scale features more intelligently. ImFPN first applies a split-attention mechanism. In this step, channels are divided into groups, and channel attention is applied within each group. This process captures global contextual information and enhances important semantic features before fusion. After the split-attention step, a similarity-based fusion module is applied. Unlike the standard FPN, which directly adds feature maps together, ImFPN computes a spatial weight map that indicates which pixels are more important during feature fusion. This allows the network to emphasize informative regions while suppressing less relevant ones, resulting in more effective multi-scale feature integration. As a result of these modifications, the average precision of the two-stage detection architecture (Faster R-CNN with ResNet-101) improved from 39.7 to 41.4.

The work regarded as our baseline for this proposal is the one done by Fang et. al. which directly addresses the problem of classification of MB in MRI by proposing the InceptionNet, which is a hybrid deep learning algorithm that integrates inception-based convolution with a self-attention mechanism, they did a binary classification of MB and non-MB using the publicly available multi-class brain tumor dataset from Kaggle and an external cohort from Shanghai Children's Medical Center, their architecture has three main components, first, a stem module applies a 3×3 convolution with 64 filters to generate low-level features from the input MRI image. Second, a modified Inception block extracts multiscale spatial features through parallel 1×1 , 3×3 , and 5×5 convolutional branches, each followed by batch normalization and ReLU activation. Instead of traditional max-pooling, a 3×3 convolution with stride 2 is used for down-sampling to better preserve spatial information. Third, a multi-head self-attention module is applied to the extracted features. This module uses four attention heads with 64-dimensional query and key vectors to compute scaled dot-product attention, enabling the model to dynamically reweight spatial locations based on their contextual relevance. A classification head with dropout is then used to produce the final MB vs. non-MB prediction.

In cross-validation on the training dataset, InceptionNet achieved an accuracy of $98.07\% \pm 0.77\%$, a recall of $96.03\% \pm 3.61\%$, and an AUC of $99.41\% \pm 0.08\%$. On the external validation set, the model reached 90.94% accuracy and an AUC of 92.79%. These results exceeded those of several standard CNN baselines, including Inception, DenseNet, VGG, MobileNet, and ResNet. Grad-CAM visualizations showed that the model tended to focus on tumor regions in the fourth ventricle and adjacent posterior fossa structures, supporting its clinical interpretability.

Despite its strong performance, the authors identify several important limitations among which are the error analysis performed which reveals that many false positives occur when non-MB tumors (such as cystic ependymomas or midline gliomas) share anatomical location or imaging characteristics with MB, and that false negatives are often associated with small lesions, weak contrast, or motion artefacts. This suggests that, even with self-attention and Inception-based multiscale convolutions, the model can still struggle in borderline or atypical cases.

Proposed Methodology

Dataset Description

The proposed training dataset is the one used by the baseline work which is the Brain Tumor MRI Dataset with 14 Classes which can be gotten from [Kaggle](#), This publicly available dataset contains MRI images spanning 14 brain tumor classes including medulloblastoma, glioma, meningioma, ependymoma, and normal brain tissue among others. For this study, following the methodology of Fang et al. (2025):

- 106 MB images are used as the positive class (after removing 25 duplicate images identified through perceptual hash comparison and radiologist verification)
- 630 non-MB images are used as the negative class (randomly selected from the remaining 13 classes, after removing 6 duplicates)
- Total: 736 images, expanded to 2944 images through augmentation
- No pixel-level annotation is required, existing class-level folder labels serve as ground truth

To check generalizability of the proposed model, it will be examined on another dataset of MRI images which can be accessed on [Kaggle](#), which also contains different classes of brain tumors including MB taken under different contrast.

Proposed Architecture: FPN-InceptionNet

We propose FPN-InceptionNet, an enhanced hybrid deep learning architecture that integrates a Feature Pyramid Network into the existing InceptionNet framework. The core modification introduces a bidirectional feature hierarchy between the Inception blocks and the self-attention module, enabling every feature level to carry both spatial precision and semantic richness before attention-based reweighting is applied.

The task remains binary image classification (MB vs. Non-MB), preserving full compatibility with the original dataset and labels. No additional annotation beyond image-level class labels is required.

Data Preprocessing

For direct comparability, all processing steps from Fang et al. [5] will remain the same. This includes a Gaussian filter with the standard deviation of 2 for noise reduction, histogram equalization for contrast enhancement, random rotation (0-45 degrees), horizontal flipping and zooming (0.8-1.2) for data augmentation.

Detailed Methodology

Inception model

The Inception based CNN enhances feature extraction by applying multiple convolutional filters of different sizes within the same layer. These multi-scale filter outputs are then concatenated along the channel dimension. This allows the network to capture features at different filter scales simultaneously, resulting in more robust and diverse feature representations.

Attention Mechanism

The attention mechanism enables a neural network to selectively focus on the most relevant parts of an input while suppressing less important information. In the proposed model, a multihead self-attention mechanism is applied after feature extraction to capture global contextual relationships across the image. Specifically, the extracted feature maps are linearly projected into three representations: Query (Q), Key (K), and Value (V). Attention scores are computed using scaled dot-product attention, where each spatial location evaluates its similarity to all other locations. These scores are normalized using a softmax function and used to weight the value vectors, allowing the network to dynamically reweight features based on their contextual importance. By employing multiple attention heads, the model learns diverse and complementary global dependencies in parallel. This mechanism enhances the network's ability to model long-range interactions and highlight tumor-relevant regions, thereby improving discriminative performance.

InceptionNet

InceptionNet is a hybrid deep learning framework that integrates an Inception-based convolutional neural network with a multi-head self-attention mechanism. The Inception module extracts multiscale spatial features through parallel convolutions (e.g., 1×1 , 3×3 , 5×5), while the self-attention component captures global contextual relationships across the image. By combining local feature extraction with global attention, the model enhances its ability to detect subtle and complex tumor patterns in MRI scans, particularly for medulloblastoma identification.

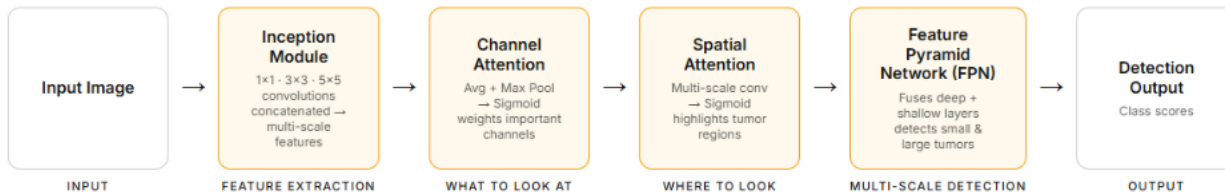
Feature Pyramid

The feature pyramid approach is our main modification to the baseline. Standard CNNs in classification pipelines typically rely on the deepest feature map only which is semantically rich but spatially coarse. As a result, small objects may be lost or blurred in deep layers. While CNNs perform well for classification tasks, since identifying what is in an image does not require

precise localization, they are less effective for object detection, which requires detailed spatial information.

To address this limitation, a Feature Pyramid Network (FPN) is incorporated. FPN is built on top of a CNN backbone and extracts feature maps from multiple layers, where each layer corresponds to a different spatial resolution. The deepest layers produce the smallest feature maps with strong semantic information, while earlier layers retain higher spatial resolution. These features are combined to form a pyramid of feature maps, enabling detection at multiple scales. This results in a rich representation that contains both semantic and spatial information.

In the context of Medulloblastoma classification, this multi-scale representation is especially important because tumors can vary significantly in size. The FPN begins with a backbone network (such as ResNet) in this case the inception-based backbone of InceptionNet that extracts feature maps at multiple levels. These feature maps are then merged and used for prediction, allowing the model to detect tumors more accurately across different scales.



FPN-InceptionNet

Our proposed methodology is the FPN-InceptionNet, an enhanced hybrid deep learning architecture that integrates a Feature Pyramid Network into the existing InceptionNet framework. InceptionNet applies self-attention to a single, downsampled feature map, which means that, as in typical CNNs, deep features are semantically strong but spatially coarse. This can make the model less sensitive to very small or weakly contrasted medulloblastomas and to subtle differences from other posterior fossa tumors. Feature Pyramid Networks were explicitly designed to address this limitation by building a top-down feature hierarchy with lateral connections, so that all scales are both semantically rich and spatially detailed (Lin et al., 2017). Incorporating an FPN into the InceptionNet backbone therefore represents a logical next step toward potentially improving robustness to lesion size and appearance variability, although its effectiveness must be validated empirically.

It is important to emphasize that this work is primarily research-oriented. The main objective is not simply to surpass the reported accuracy of InceptionNet, but to systematically investigate how explicit hierarchical multi-scale representations, via an FPN, interact with Inception-style convolutions and self-attention in the context of medulloblastoma classification. Even if

performance gains prove to be modest, the study aims to provide insight into when and how feature pyramids are beneficial for posterior fossa tumor imaging, to clarify their limitations on relatively small medical datasets, and to contribute empirical evidence that can inform the design of future architectures.

Hyperparameter

For fair comparison with the baseline, the following hyperparameters will be used.

Hyperparameter	Value
Optimizer	Adam
Initial Learning Rate	0.005
Learning Rate Schedule	Reduce on plateau
Batch Size	8
Maximum Epochs	40
Early Stopping Patience	10 epochs
Dropout Rate	0.3
Cross-Validation	5-fold

Evaluation Metrics

All evaluation metrics from Fang et al. (2025) are retained:

Accuracy, Precision, Recall, F1-score, AUC, ROC curve

Results reported as mean $\pm$ standard deviation across 5 folds

External validation on the proposed Kaggle dataset for validation.

Interpretability

Because healthcare is a sensitive domain and interpretability is highly important, Grad-CAM heatmaps will be generated at each pyramid level for enhanced interpretability.

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